



Voltage Drop Analysis Report

Voltage Drop Calculation Report

Project: Office Building A

Project Number: P-001

Client: ABC Construction

Engineer: Mafizul Islam

Date: 20260611

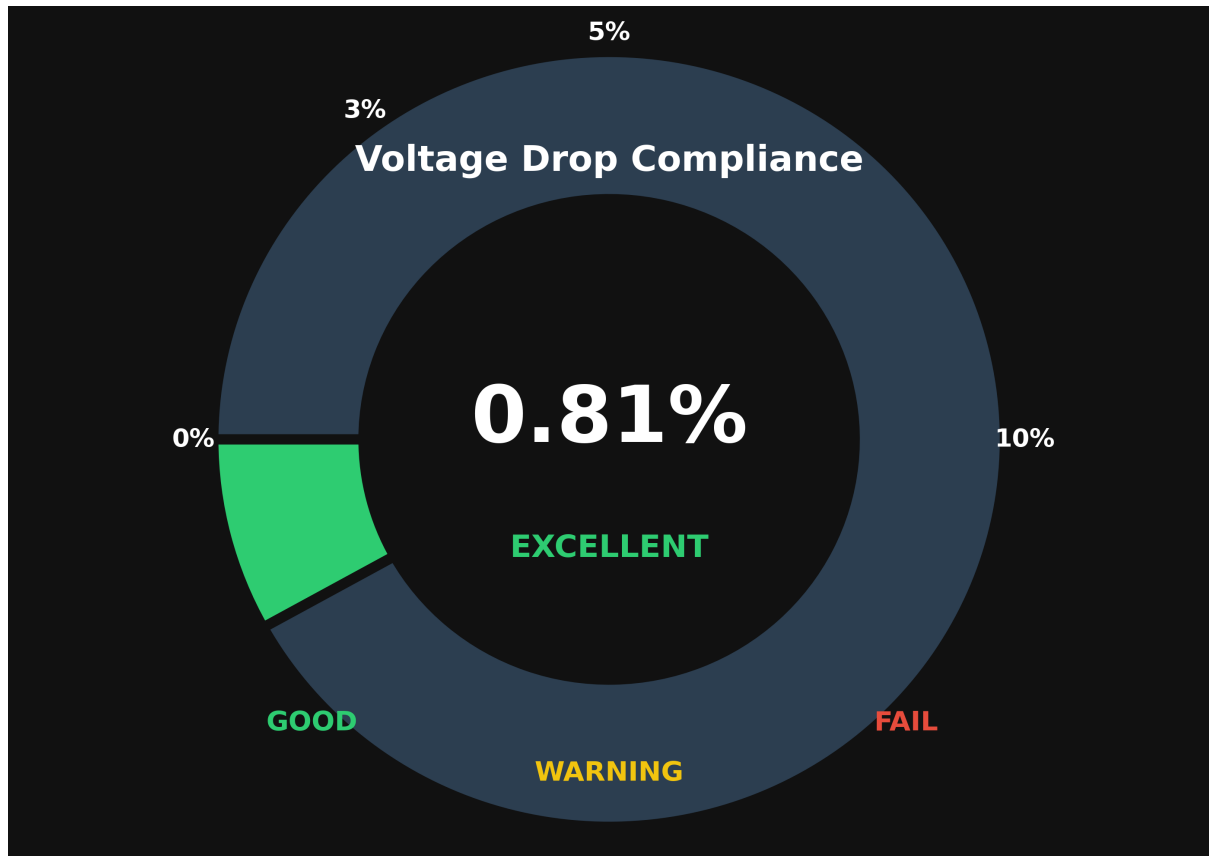
Calculation Summary

PROJECT STATUS
Voltage Drop: 0.81%
IEC Limit: 5.00%
Overall Result: GOOD
Recommended Cable: 10 mm²
Recommended Breaker: 63 A
TRANSFORMER ANALYSIS
Calculated Load: 35.56 kVA
Design Load (25% margin): 44.44 kVA
Selected Transformer: 100 kVA
Transformer Loading: 35.6%
Loading Status: GOOD RESERVE CAPACITY

Parameter	Value
Power (kW)	32.0
Voltage (V)	400.0
Power Factor	0.9
Length (m)	50.0
Phase	Three Phase
Cable Material	Copper
Installation Method	Conduit
Current (A)	51.32
Voltage Drop (V)	3.23
Voltage Drop (%)	0.81%
Recommended Cable	10 mm²
Recommended Breaker	63 A

Parameter	Value
Status	GOOD
Design Check	Breaker (63A) is undersized for load current (51.3A). Consider next higher size.

Voltage Drop Compliance Gauge



Engineering Recommendation

Excellent design margin available. Voltage drop is significantly below IEC limit.

Engineering Calculations

Current Calculation

Formula: $I = P \times 1000 / (\sqrt{3} \times V \times PF)$

Substitution: $I = 32.0 \times 1000 / (\sqrt{3} \times 400.0 \times 0.9)$

Result: $I = 51.32 \text{ A}$

Voltage Drop Calculation

Formula: $VD = 1.732 \times I \times R \times L / 1000$

Result: $VD = 3.23 \text{ V}$

Voltage Drop Percentage: 0.81%

Engineering Notes

- Voltage drop should normally be less than 5% for main circuits.
- Cable sizing recommendation is indicative only.
- Final cable sizing shall be verified according to full IEC standards.
- Installation method, ambient temperature and grouping factors must be checked.
- Final design approval remains the responsibility of the engineer.